

Week 1: Review of JSS 2 Work

JSS3 Intermediate Science — First Term Lesson Note (Deep-Dive Edition)

Introduction

As we begin JSS3, this week reviews essential concepts from JSS2, reinforces laboratory safety, and previews what this year's Intermediate Science course will cover.

Revision: Matter

Matter is anything that has mass and occupies space. Matter exists in three main states: **solid** (fixed shape and volume), **liquid** (fixed volume, takes the shape of its container), and **gas** (no fixed shape or volume, fills its container). Matter can change between these states through processes such as melting, freezing, evaporation, and condensation.

Revision: Energy

Energy is the ability to do work. Common forms include **kinetic energy** (energy of motion), **potential energy** (stored energy), **heat energy**, **light energy**, **chemical energy**, and **electrical energy**. The Law of Conservation of Energy states that energy cannot be created or destroyed, only changed from one form to another.

Revision: Classification

Classification is the grouping of things based on shared characteristics. Living things are classified based on features such as cell structure, method of feeding, and reproduction, while materials can be classified as elements, compounds, or mixtures.

Laboratory Conduct and Safety

- Always wear appropriate protective equipment (lab coat, goggles) when instructed.
- Never taste, smell directly, or touch unknown chemicals.
- Report all spills, breakages, or injuries to your teacher immediately.
- Keep your workspace clean and free of unnecessary bags or materials.
- Know the location of safety equipment: fire extinguisher, first aid kit, and emergency exits.
- Never carry out unauthorized experiments.

Introduction to JSS3 Concepts

This term's Intermediate Science course focuses on **environmental science** — ecosystems, natural resources, environmental hazards, environmental health, waste management, and sustainable development. Later terms will build on this with cell biology, chemistry, and further physical science topics.

Real-Life Application

Understanding states of matter helps explain everyday phenomena like why ice melts in a warm room, while laboratory safety rules protect students during hands-on experiments throughout the year.

Vocabulary

Word	Meaning
Matter	Anything that has mass and occupies space
Conservation of Energy	The principle that energy cannot be created or destroyed, only transformed
Classification	The grouping of things based on shared characteristics

Evaluation Questions

- 1 Define matter and name its three main states.
- 2 State the Law of Conservation of Energy.
- 3 Give three examples of forms of energy.
- 4 Explain the difference between an element, a compound, and a mixture.
- 5 List three important laboratory safety rules.
- 6 Why should unknown chemicals never be tasted or smelled directly?
- 7 What should you do if you spill a chemical in the laboratory?
- 8 Name the broad theme of this term's Intermediate Science topics.
- 9 Explain the difference between kinetic and potential energy, with an example of each.
- 10 In your own words, explain why laboratory safety is important for the whole class, not just one student.